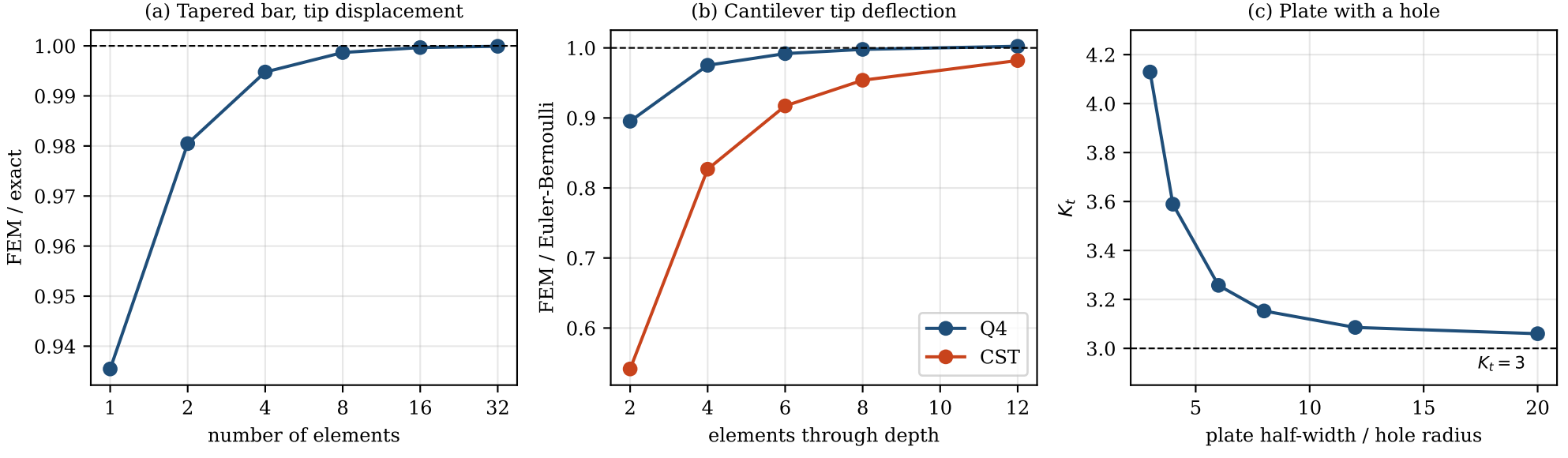
(a) Tapered bar, tip displacement
FEM / exact
1.00
0.99
0.98
0.97
0.96
0.95
0.94
1
2
4
8
16
32
number of elements
(b) Cantilever tip deflection
FEM / Euler-Bernoulli
1.0
0.9
0.8
0.7
0.6
2
4
6
8
10
12
elements through depth
Q4
CST
(c) Plate with a hole
K_t
4.2
4.0
3.8
3.6
3.4
3.2
3.0
$K_t = 3$
5
10
15
20
plate half-width / hole radius